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A computationally efficient multiphysics model for friction stir welding with polygonal pin profiles

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ABSTRACT

This study aims to model temperature distribution in friction stir welding (FSW) using various backing plates and polygonal pin profiles since temperature significantly modifies the microstructure and texture of the weld zone affecting the weld quality such as weld strength, hardness, etc. The experimental results depict the importance of temperature on the grain size and tensile strength of the materials. However, determining the temperature at each point of the weld is difficult and expensive in the case of experiments. Therefore, in order to accomplish the objective, it is necessary to perform simulations. This paper presents a 3-D transient multiphysics model developed for FSW combining multiple physical phenomena such as heat transfer and structural mechanics in a unified framework, COMSOL. A viscoplasticity model is chosen as behavior for the AA1100 aluminum material using Anand viscoplasticity. It is computationally efficient and accurate. The model fidelity to the twin FSW process is achieved by considering temperature-dependent yield strength. Modeling results show the polygonal pin profile edges to be influencing the temperature. Increasing the number of faces on the pin sides leads to a higher temperature. Specifically, transitioning from an octagonal to a decagonal profile results in a minimal increase in total heat generation. As the pin shape approaches cylindrical, there is a gradual convergence in heat generation with that of a cylindrical pin. Experiments are also carried out that validate simulation results. Overall, the model is sufficient to twin the process for predicting weld quality and is Industry 4.0-compliant.

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1. Introduction

FSW has been extensively applied in various applications such as automobiles, aerospace, shipbuilding, railways, etc. [1–3]. It is helpful in welding materials with higher strength-to-weight ratios [4], such as aluminum and magnesium alloys [5],

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contributing to vehicle mass reduction by improving fuel economy and lowering overall energy usage [6]. However, because of the process complexity, the level of difficulty increases in gaining an understanding of weld profile, microstructure, and mechanical properties during FSW [7–9]. To address this issue, the application of numerical simulation has demonstrated its effectiveness as a valuable technique for investigating the development of FSW joints [10]. Numerous approaches such as Finite Element Method (FEM), Arbitrary Lagrangian-Eulerian (ALE) Method, Eulerian approach, Smoothed Particle Hydrodynamics (SPH) method, Coupled Eulerian-Lagrangian (CEL) method, Material Point Method (MPM) approach, etc. have been employed for FSW to replicate and analyze the joining procedure utilizing various commercial finite element (FE) software, such as ABAQUS, ANSYS, DEFORM 3D, and others [11–13].

The stirring zone in FSW undergoes significant deformations and strain rates as a result of the intricate interplay between heat transfer and solid mechanics within the materials being welded and the factors are rotation of the welding tool, frictional heating, plastic deformation, high strain rates, severe shear deformation, etc. [14]. An issue in the models developed in a few earlier works is that they have not considered the plastic deformation happening in the FSW process [15–17]. Heat transfer calculations were done by using analytical models without considering the plastic deformation of the material near the tool. The researchers suggest simulating heat transfer using two boundary value problems: (a) transient heat transfer for workpieces and, (b) steady-state heat transfer for the tool. The reason behind this is since the workpiece and the tool may experience different time scales of temperature changes during FSW, employing separate boundary value problems allows for a more accurate representation of the thermal behavior of each component. Workpieces and tools have different geometries, compositions, and contact conditions, resulting in variations in heat dissipation. The transient heat transfer analysis for workpieces takes into account factors such as heat generation, conduction, convection, and radiation, which may vary over time. In contrast, the steady-state heat transfer analysis for the tool assumes a constant heat dissipation rate, which can simplify the modeling process while still providing reasonable accuracy. The approach is reasonable for determining frictional heat generation. However, it is silent about the plastic deformation phenomenon which also contributes to heat generation. Further, the temperature evolution was used to calculate residual stresses across the welding domain without considering plastic deformation [15]. An earlier work built a 3-dimensional heat flow model based on the finite difference method (FDM) for FSW [16]. The generation of heat was assumed to occur at the tool shoulder only because of friction. The coefficient of friction was tailored to maintain the maximum temperature beneath the melting point of workpieces. The model applicability is limited because heat generation due to tool pin and plastic deformation was not considered. Further, a 3-dimensional thermal model was also developed for FSW [17]. The convective heat transfer coefficient of the bottom side of the plates was modified to assuring a good agreement between simulated and observed temperature values. Another issue with the numerical models developed for FSW is that the temperature-dependent properties were not considered. A 3-dimensional viscoplastic model was developed to study the temperature profiles

in an aluminum alloy (AA 7050-T7451) [18]. The simulated temperature values were reported to be over-predicting the experimental measurements. This inconsistency was probably due to inadequate representation of the constitutive behavior of the material, over a broad range of strain rates, temperatures, and stresses commonly encountered in FSW. Though there is a good agreement between measured and simulated results, the temperature-dependent properties of the material were not considered, which further added to the model inconsistency. A more enhanced model than [10] was developed by considering the viscoplastic material flow near the tool [19]. The heat flux was calculated with functions of coefficient of friction and slip rate. Despite the improvement in accuracy, the heat generation rate was assumed to be constant during the welding stage. The assumption limits the potential of the model. In addition, the heat generation from the bottom portion of the pin was also neglected, and a backing plate for proper heat conduction was not used. A work studied the influence of material models by using an Eulerian model developed for FSW in [20]. The variations in temperature values were found to be more with Kocks and Meckin material model and less with the Johnson-Cook (JC) material model. The possible reason may be the formulation and set of parameters that govern its behavior under different conditions. The possible reason for this result is that the Kocks-Mecking model focuses on the strain-hardening behavior of crystalline materials, with a specific emphasis on dislocation interactions. It considers factors such as dislocation density and velocity to describe the response of the material to plastic deformation while the JC material model accounts for the dynamic behavior of materials under high strain rates and elevated temperatures incorporating parameters such as strain rate, temperature, and strain to characterize the response of the material, including strain hardening, strain rate sensitivity, and thermal softening.

A thermo-mechanical model for FSW was also developed considering both viscous dissipation and frictional heating into account [21]. The material flow patterns in the neighborhood of the pin were shown using the particle tracing method. However, the temperature-dependent yield behavior of the material was not considered. Currently, FSW is not limited to aluminum alloys [22, 23]. Hence, while modeling FSW with different materials such as high-strength steel and others [8, 24–27] etc., the temperature-dependent yield value must be considered during modeling to get accurate results. Recent developments in modeling FSW were shown using the Coupled Eulerian-Lagrangian method where the mass scaling method was applied to minimize the computational time [28]. It is worth working with software having user-friendly interfaces to run virtual FSW tests with numerical modeling which will reduce the cost of experimentation. Strong knowledge of the heat transfer process from the model may aid in anticipating the thermal cycles during welding, hardness in the weld zone, microstructural evolution assessing the weld quality in terms of ultimate tensile strength, computing the temperature-dependent viscosity while modeling material flow, etc. [29].

The extensive discussion of the published literature suggests that a lot of research has been directed toward the development of physics-based models for FSW. However, those models cannot replicate the complete physics as they overlook all sources of heat generation and plastic deformation during modeling. Their

applicability is also limited because temperature-dependent properties of the workpiece were not considered, particularly, temperature-dependent yield strength. When the temperature reaches the solidus temperature (T_solidus) of the plate, the yield strength of the material decreases. Therefore, the heat generation is reduced near the tool and workpiece interface when it exceeded the T_solidus of the material and thus permits the plates to regain their yield strength. At temperatures above the solidus point, a material behaves as a liquid. Therefore, the material close to the tool/matrix junction produces very little heat if it exceeds the T_solidus. This reduces the temperature level and enables the material to regain its strength. Addressing these limitations is crucial to develop an accurate and robust simulation twin of FSW.

To address the discussed problems, a multiphysics model [30] is developed for FSW that considers temperature-dependent yield strength, plastic deformation, and all possible sources of heat generation i.e. due to shoulder, pin periphery, and bottom of the pin. All these attributes are fulfilled in one model which is developed using COMSOL to make it suitable for industrial applications. It reduced the complexity of the model, making it flexible and easy to use. Further, the developed model was applied to study the process when welding with polygonal pin profiles as there is less research being done on designing FSW tool pins [22, 31]. Detailed numerical investigations are carried out to analyze the effects of the thermal field during welding with different backing plates which is described in Section 3.2.5 and offer guidance on selecting the preferred choice of the backing plate for aluminum AA1100 welding. The current model is also compared with models in the existing literature which shows the computational effectiveness of the current model.

2. Experimental and simulation details

In FSW, at the tool-workpiece interface, heat is generated due to friction and plastic deformation of the material. Heat transfer and solid mechanics are the main physics involved during FSW. For this current work, the welding was carried out using a linear numerically controlled FSW machine created by ETA Bangalore Pvt. Ltd. with an in-house experimental setup. The length and breadth of the AA1100 aluminum alloys workpiece plates were both 84 mm and 38 mm, respectively, having 3 mm thickness. H13 stainless steel was used as the tool material with the cylindrical shape pin with a shoulder diameter of 16 mm, and a pin diameter of 5 mm, with a pin height of 2.8 mm. The plunge depth considered for all the experiments was 0.1 mm. The experiments were performed with three different tool rotational speed (ω) values of 600 rpm, 900 rpm, and 1200 rpm at 0.1 mm plunge depth (PD) and a constant welding speed (v) of 100 mm/min. To acquire temperature data, K-type thermocouples were used with an NI data acquisition card. The thermocouples were inserted in a 1 mm blind hole at a distance of 1 mm from the end of the tool shoulder and in the middle of workpiece. The complete experimental setup was depicted in Figure 1.

COMSOL software is having a user-friendly interface and can create multiphysics models to solve various partial differential equations simultaneously related to different physics in one fully coupled system [32]. In this current study, a 3D transient thermo-mechanical numerical model is developed using COMSOL® Software Version



Figure 1. Experimental setup.

5.6 with heat transfer, structural mechanics, and nonlinear structural materials modules to model the FSW process. A viscoplasticity model assumption is chosen as behavior for the AA1100 aluminum material of the assembly. Multiphysics coupling was done for thermal-structural interaction by considering thermal expansion and temperature coupling to source from the heat transfer module solution to the solid mechanics solution. The complete modeling details are given below.

2.1. Model geometry, meshing, and process parameters

The workpiece, tool dimensions, and materials for simulations are in-line with the experiments. The tool was considered as a rotating circular moving heat source in the model considering that it was already inside the workpieces. In this model, the moving heat source with radius R was supposed to travel (same speed as the tool) in the X-direction to accomplish the welding. The heat source represents the heat generation due to the shoulder surface, the boundary surface of the pin, and the bottom surface of the pin. The heat fluxes were prescribed using mathematical descriptions. The variable for the radial coordinate R is given by Equation (1).

$$R = \sqrt{(\mathrm{x} - v\mathrm{t})^2 + \mathrm{y}^2} \tag{1}$$

The heat flux due to the tool shoulder will be accommodated in the particular zone when the R value will be less than and equal to the shoulder radius, and the rest of the weld zones will experience zero flux. The complete geometrical model is shown in Figure 2 (a). For heat conduction, a backing plate of cast iron with $84*64*5\,\mathrm{mm}^3$ dimension was considered on the bottom side.

A mesh sensitivity analysis was used to find the best mesh size for the plate. The accuracy and computation time are strongly influenced by the mesh size. While a coarser mesh results in erroneous results, a finer one takes longer to compute. As a result, a mesh sensitivity analysis was performed to find the optimal balance between accuracy and computing time. In this analysis mesh size of the workpiece was chosen

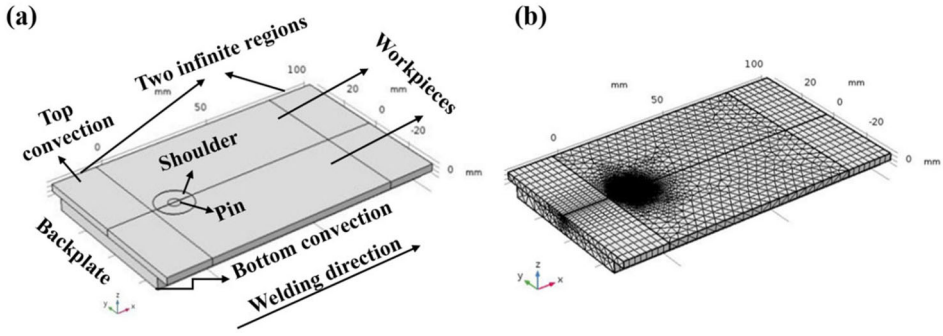


Figure 2. (a) Geometrical model and (b) meshing.

Table 1. Mesh sensitivity analysis.

Mesh type	Mesh size (mm)	Number of elements	Solution time (h: min)	Maximum temperature (°C)
Extremely fine	0.025	1056689	39:21	306.95
Extra fine	0.188	251109	17:36	306.25
Finer	0.5	45407	3:05	306.65
Fine	1.25	13356	2:10	302.45
Normal	2.25	9077	1:28	294.96
Coarse	3.5	5390	00:59	296.72
Coarser	5	3615	00:41	286.2
Extra coarse	6.75	1901	00:36	284.59
Extremely coarse	8.75	1298	00:24	291.36

from extremely coarse to extremely fine mesh (i.e. from 8.75 mm to 0.025 mm) for a set of parameters ($\omega = 1200$ rpm, $v = 100$ mm/min, PD $= 0.1$ mm), and the output response (the maximum temperature at 1 mm distance from the tip of the shoulder of FSW tool and just exactly in the middle of the workpiece) was compared for all mesh sizes. Table 1 shows the impact of mesh size on the computation time and maximum temperature in the welding phase of FSW. The configuration of the computer used for the simulation was Intel®, Xenon® Gold 6238 R CPU@2.20 GHz processor (2 processors with 28 cores each) with 256 GB RAM.

It took 39 h 21 min and 00 h 24 min, respectively, to compute the extremely fine (0.025 mm) and extremely coarse (8.75 mm) meshes.

The increased number of elements in a finer mesh leads to an increase in computing time. Because of this, the time it takes to solve the element stiffness matrix rises. It is observed that mesh size has not much effect on the maximum temperature from extremely fine mesh type to finer mesh type as depicted in Table 1. The difference in maximum temperature between mesh sizes of finer (0.5 mm) and extra fine (0.188 mm) is found to be less than 1%. In other words, with further mesh refinement, the maximum temperature stays the same. As a result, the mesh size of 0.5 mm looks to be the best value for meshing at the interaction region of the workpiece and tool for this current model. A mesh size of 6.8 mm was used for the rest of the region in order to save calculation time as shown in Figure 2 (b).

Table 2. Temperature-dependent yield criterion of pure aluminum AA1100.

Temperature (°C)	371	316	260	204	149	100	24
Yield strength (MPa)	11	14	18	24	29	32	34

2.2. Novelty and assumptions

A novel concept of temperature-dependent yield criterion of pure aluminum AA1100 was considered in the current model, and the values are tabulated in Table 2. The yield stress decreases as the temperature of the plates approaches the solidus temperature (T_solidus). Hence, heat generation is minimized around the tool-workpiece contact when it exceeds the T_solidus of material, allowing the plates to restore their yield strength. This effect, which happens below the T_solidus, is called a self-stabilizing effect. This makes the model more sensitive to rely on changes in boundary conditions and welding parameters [33]. Therefore, it is crucial to include a temperature-dependent yield criterion in the model.

The effect of temperature on AA1100 properties (and thus its yield stress) can be accurately described *via* an interpolation function based on experimental data, which is tabulated in Table 2. The importance of temperature-dependent yield strength is also shown in Section 3.2.2.

An additional aspect of novelty in this study is the accurate calculation of overall heat generation from the shoulder surface, pin periphery, and bottom surface of the pin for various pin profiles. These equations can be used to analyze further the relationship between the temperature distribution and weld quality i.e. ultimate tensile strength using polygonal pin profiles. Another unique aspect of this research is the computational efficiency of the current model, which has been validated through a comparison with existing models in the literature.

The following assumptions were considered in the present work:

- Uniform contact frictional shear stress (τ) was acknowledged throughout the model
- The tool tilt angle was not considered
- Plunging, dwelling, and retract cycles were not considered
- When the local temperature approaches the T_{melt}, no heat flows into the workpieces.

2.3. Thermal model

The 3-D heat conduction equation which governs the thermal analysis is presented in Equation (2):

$$\rho C_p \frac{\partial T}{\partial t} + \left(-k\nabla^2 T\right) = \dot{q} \tag{2}$$

where, ρ, C_p, k are density, specific heat capacity, and thermal conductivity.

2.3.1. Generation of heat due to shoulder surface

Heat generation which ensues within the workpiece-interface and tool shoulder is expressed in Equation (3) [34]. It is indicated with local heat flux per unit area (W/m^2). It was applied within a distance R from the central tool axis.

$$q_{shoulder\ \ surface}\ (R,\ \ T) = \left\{ \left(\frac{2}{3} \times \pi \times \omega \times \tau_{contact} \times \left(r^3_{shoulder} - r^3_{pin}\right),\ \ if\ \ T < T_{melt}\ \ 0\ \ if\ \ T \geq T_{melt}\right\} \right. \tag{3}$$

where contact shear stress $= \tau_{contact} = \delta \times \frac{\sigma_{yield}}{\sqrt{3}} + \left(1 - \delta \times \mu \frac{F_n}{A_s}\right)$, F_n refers to the normal force, μ is the coefficient of friction, A_s denotes shoulder surface area, $\delta =$ Slip rate, σ_{yield} is yield shear stress [34]. When the temperature is equal to or more than (T_{melt}), the model sets the heat generation to zero.

2.3.2. Heat generation due to probe surface

The model simulates the heat generation between workpieces and pin boundary surfaces as a boundary heat source, which is given in Equation (4) [34].

$$q_{pin\ side}(T) = 2\ \ \times \pi \times \tau_{contact}\ \ \omega \times r^2_{pin} \times h_{pin} \tag{4}$$

Here, h_{pin} refers to the height of the pin, r_{pin} *is the radius of the pin*, ω is the angular velocity of the tool (rad/s).

2.3.3. Heat generation due to the bottom of the probe

Heat generation which ensues within the workpiece-interface and probe bottom surface, is given in Equation (5) [34].

$$q_{pin\ bottom}\ (R,\ \ T) = \left\{ \begin{matrix} \frac{2}{3} \times \pi \times \tau_{contact} \times \omega \times r^3_{pin}, & if\ \ T < T_{melt} \\ 0 & if\ \ T \geq T_{melt} \end{matrix} \right\} \tag{5}$$

2.4. Anand viscoplasticity material model

Anand viscoplasticity model [35] was adopted for strain rate dependency on stress in the workpiece. This model was made for aluminum alloys with viscoplastic behavior, and it can be used in engineering in a useful way [36]. The model was made in such a way that it could be used to study the rate-dependent deformation of metals when they are heated to high temperatures.

The single-scalar internal variable model provides a straightforward set of constitutive equations for minor elastic deformations and large isotropic viscoplastic deformations. The temperature sensitivity, strain hardening, strain-rate history, and dynamic recovery are all accounted for in the constitutive equations. To capture the isotropic resistance to plastic flow provided by the internal state of the material, this model uses a single scalar as an internal variable (S). It stands for the average isotropic resistance to macroscopic plastic flow provided by the underlying isotropic

strengthening processes such as sub-grain and grain size effects, solid solution strengthening, dislocation density, etc. For this reason, S is consequently proportional to σ which is defined as

$$\sigma = C.S; \;\; C < 0, \tag{6}$$

where C is a material parameter and constant in the constant strain rate test, which is defined as:

$$C = \frac{1}{\xi}\sinh^{-1}\left[\frac{\dot{\varepsilon_p}}{A}\exp\left(-\frac{Q}{RT}\right)\right]^m \tag{7}$$

The model was first made for applications that involved forming operations. It can also be used to solve viscosity problems, which involve strain rate and temperature. FSW is one of these processes where at high temperatures, the effects of temperature magnitude and history, strain rate, and strain hardening have a big effect on materials. Hence, the selection of Anand viscoplasticity material model will be of great significance in the context of FSW. By choosing this material model, it becomes possible to accurately represent the behavior of the materials involved in the FSW process. It is a constitutive equation used for small elastic deformation, and large, isotropic, viscoplastic deformation. The flow equation is highlighted in Equation (8) [37].

$$\dot{\varepsilon} = A \;\exp\left(-\frac{Q}{RT}\right)\left[\sinh\left(\xi\frac{\sigma}{S}\right)\right]^{\frac{1}{m}} \tag{8}$$

In the flow equation, the internal state variable is introduced only as a ratio with the equivalent stress. A classical Arrhenius term is used to incorporate the temperature dependence in Equation (8). The evolution equation for the internal variable $\dot{S}$ is assumed to be of the form:

$$\dot{S} = g(\sigma, S, T)\dot{\varepsilon_p} \tag{9}$$

where the function g (σ, s, T) is associated with dynamic processes that are strain hardening and dynamic recovery. A simple form of evolution of Equation (9) was given by Anand as follows in Equation (10):

$$\dot{S} = \left[h_0\left|1 - \frac{S}{S^*}\right|^a .sign\left(1 - \frac{S}{S^*}\right)\right].\dot{\varepsilon_p}; \;\; a > 1, \tag{10}$$

$$S^* = \hat{S}\left[\frac{\dot{\varepsilon_p}}{A}\exp\left(\frac{Q}{RT}\right)\right]^n \tag{11}$$

The commercial finite element software COMSOL already has the Anand viscoplastic model built in, which makes it much easier to use. The material parameters

Table 3. Anand Viscoplasticity model parameters for AA1100.

Symbols	Model parameters	Values
A	Viscoplastic rate coefficient	1.91e + 7 1/s
Q	Activation energy	175300 J/mol
ξ	Stress multiplier	7
R	Universal gas constant	8.314 J K^{-1} mol^{-1}
M	Stress sensitivity	0.23348
S_0	Deformation resistance saturation coefficient	18.9e + 9 N/m^2
S_{init}	Deformation resistance initial value	36.6e + 9 N/m^2
h_0	Hardening constant	1115.6e + 9 N/m^2
A	Hardening sensitivity	1.3
N	Deformation resistance sensitivity	0.07049

and their values are given in Table 3. All the Anand viscoplasticity material parameters for aluminum AA1100 are taken from the reference [37].

The model has been implemented in COMSOL, due to the ease with which simulations across multiple physics domains can be coupled. After the completion of creating the proposed model, researchers can automate it through MATLAB, which would be switched with automation *via* Java code. Thus, the entire work can be encapsulated in an easy-to-use interface within COMSOL, as a COMSOL application. To work with different material models, through the API, COMSOL Multiphysics modeling can be controlled using MATLAB scripts (LiveLink^{TM} for MATLAB). Furthermore, a COMSOL simulation application can contain MATLAB calls, in which MATLAB can start automatically during run-time.

2.5. Boundary conditions

- The velocity of the plate side surfaces is constrained in three directions (X, Y, and Z).
- The tool is considered a heat source for the shoulder and pin.
- Two infinite regions in the X-direction surround each plate as depicted in Figure 2 (a). Here, the material exits the computational domain by considering the temperature to be in equilibrium with the atmosphere.
- Heat loss is assumed due to the convection and radiation from the upper and side surfaces of the workpiece, as well as from the lower side of the backplate. The heat flux expressions for the above-mentioned surfaces are given in Equations (12,13), respectively [38].

$$q_{upper} = h_u(T_0 - T) + \varepsilon\sigma\left({T_p}^4 - T^4\right) \tag{12}$$

$$q_{bottom} = h_d(T_0 - T) + \varepsilon\sigma\left({T_p}^4 - T^4\right) \tag{13}$$

where h_u and h_d are heat transfer coefficients for natural convection. Other notations and their respective values are tabulated in Table 4.

Table 4. Parameters used in simulations and their values.

Notation	Parameter	Value
E	Emissivity	0.4
T_0	Reference temperature	293.15 K
Σ	Stefan-Boltzmann constant	5.67×10^{-8} Kg s^{-3} K^{-4}
T_p	Temperature of ambient air	300 K

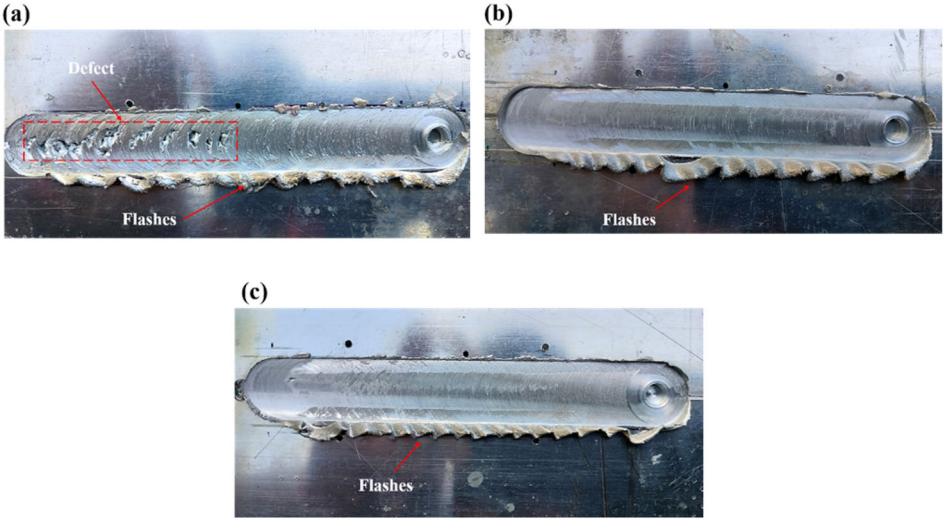


Figure 3. Welded samples with ω of (a) 600 rpm (b) 900 rpm (c) 1200 rpm at a constant v of 100 mm/min, PD of 0.1 mm.

3. Results and discussion

The following section is subdivided into two parts: Experimental findings and simulation outcomes.

3.1. Experimental findings

3.1.1. Weld appearance

Three welded samples were joined using three distinct process parameters: three values of rotational speed (ω), namely 600, 900, and 1200 rpm. The welding speed and plunge depth were constant at 100 mm/min and 0.1 mm, respectively as stated in section 2. Figure 3 displays the visual appearance of the three welded samples. It can be observed that at lower ω i.e. at 600 rpm, the formation of defects was caused because of less amount of heat generation resulting in improper mixing of the material. The roughness value was measured with a 3D non-contact profilometer (CCI MP, Taylor Hobson, UK) using an objective lens of 20x and a sampling area of 0.84×0.84 mm on the center of the weld joints. The roughness values obtained are 8.778 µm, 3.458 µm, and 2.813 µm for ω i.e. 600, 900, and 1200 rpm, respectively, with a constant v of 100 mm/min and PD of 0.1 mm.

An enhancement in surface smoothness becomes evident as the rotational speed (ω) increases. Higher ω results in increased heat generation and deformation in the

weld zone, facilitating uniform material mixing and leading to improved surface finish. Additionally, it is observed that higher ω values lead to a reduction in the presence of flashes. In the FSW process, it is essential to ensure adequate thermal softening and forging effects in the materials. This requires the shoulder of the welding tool to be pressed into the workpiece to a sufficient depth. The extruded material forms the flashes of the weld, and the joint exhibits a noticeable depression compared to the base metal, resulting in the phenomenon of weld thinning. The extent of weld thinning is defined as the difference between the thickness of the base material and the thickness of the weld joint. The weld joint with tool rotational speed of 600, 900, and 1200 rpm has a weld thinning of 0.35, 0.20, and 0.18 mm, respectively, which is 11.66%, 6.66%, and 6% of the base metal thickness. The reason for more thinning in lower ω may be attributed to a decrease in temperature which leads to improper mixing of material and in turn was reflected in the form of flash and defects and excessive thinning of the weld zone. It also affected the tensile properties which is depicted in Figure 8 (a).

3.1.2. Microstructural evolution

To analyze the microstructure, the AA1100 base metal, and welded samples were prepared for observation. Initially, the samples were cut using a sample cutting machine (Struers, Discotom) and then polished using silicon-carbide emery papers with different grit sizes (ranging from 280 to 2000 µm). Following this, diamond cloth polishing was carried out using a semi-automatic sample grinder and polisher (Struers, LaboPol). For microstructural analysis of the welded samples, the Electron Backscatter Diffraction (EBSD) technique was employed. The cut samples were subjected to electro-polishing in an A2 electrolyte solution at 40 V for a duration of 30 s. The EBSD scan was performed on stir zone (SZ) using a field emission gun scanning electron microscope (ZEISS, AURIGA) equipped with an Oxford detector. The scan was conducted at a scan rate of 0.5 step size. To process the data obtained from EBSD, software (TSLOIM Analysis 7.0) was utilized. The data were cleaned and analyzed using the standard grains dilation method in a single iteration.

The microstructure of the base material (BM) and SZ of welded samples have been depicted in Figure 4. It can be observed that the BM has the bi-model structure of grains. The presence of neckless-shaped finer grains in the BM is because of the deformation produced in the sheet-making process. The formation of coarser grain in base material is likely due to the manufacturing processes involved, such as rolling. The BM was not in annealed state; therefore, some finer grains were also seen near the coarser grains. The average grain size of BM was found to be 15.77 μm. Refinement in the microstructure occurs at the SZ because of the stirring action of the tool in FSW. The average grain size at SZ of the welded samples were 4.41, 3.70, and 2.70 μm with the ω of 600, 900, and 1200 rpm, respectively. Figure 4 illustrates the distribution of grain sizes and the average diameter, with the merged data shown in the right corner. The calculation of grain size is performed using an area-weighted averaging method. This approach involves a straightforward computation which can be expressed as follows: $d_{area} = \frac{1}{\sum_{i=0}^{n} A_i} \sum_{k=0}^{n} A_i d_i|$, where n represents the number of grains, A_i denotes the area of grain i, and d_i signifies the diameter. The advantage of

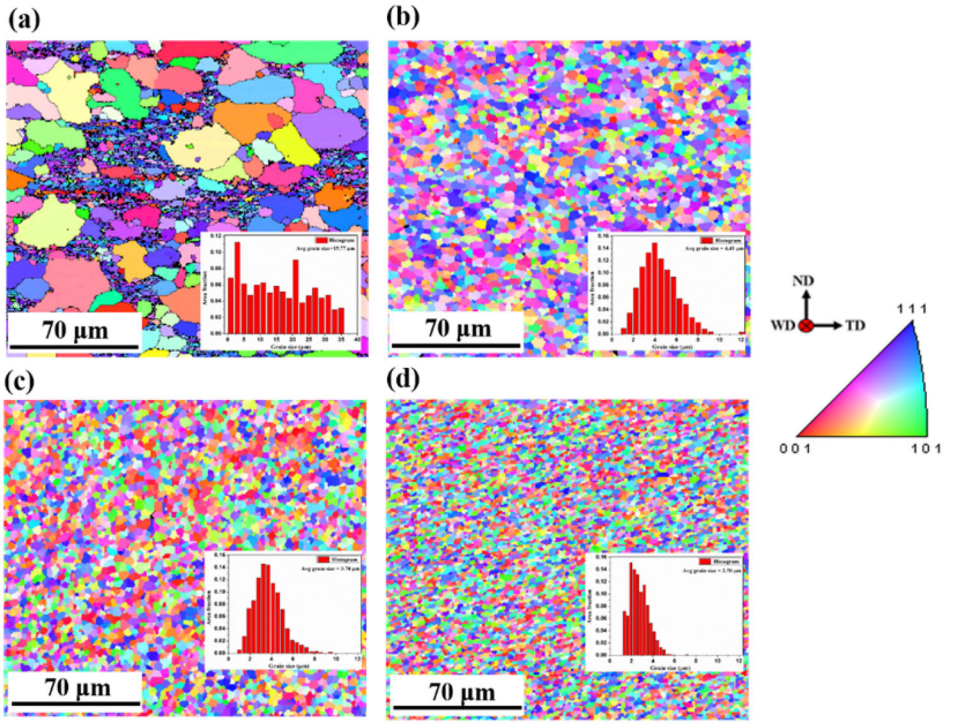


Figure 4. EBSD OIM plots of (a) BM, and SZ of welded samples with Varying ω (b) 600 rpm (c) 900 rpm (d) 1200 rpm.

the area average, compared to determining it through curve fitting, lies in the fact that it is derived directly from the original grain size data, irrespective of the binning method employed to construct the distribution plot.

It was found that the grain size was to be dependent on the process parameters. Finer grains lead to an increase in the joint strength of the weld which was described in Section 3.1.3. Smaller grains exhibit higher surface area-to-volume ratios, resulting in an increased proportion of grain boundary to dislocations. An increase in the number of grain boundaries directly contributes to enhanced strength. In the current study, finer grains were obtained with ω of 1200 rpm. The Hall-Petch relationship established a correlation between the yield stress and the size of grains [39]. Therefore, numerous studies have successfully applied the Hall-Petch relationship to the ultimate tensile strength, demonstrating a strong correspondence between these two, and also in this study, the relation satisfies well as depicted in Figure 8 (b).

The formation of grain structure in FSW is due to dynamic recrystallization [40]. Grain orientation spread (GOS) maps shown in Figure 5 describe the extent of deformation of the grains.

The grains will be considered as recrystallized grains if the GOS value comes less than 1^0, and the rest are considered as deformed grains. The fraction of recrystallized grains depicted in blue color were nearly 70, 74, and 80% for ω of 600 rpm, 900 rpm, and 1200 rpm respectively as shown in the GOS maps in Figure 5 (a–c). The colorful grains excluding the black spot portions are the recrystallized grains as depicted in

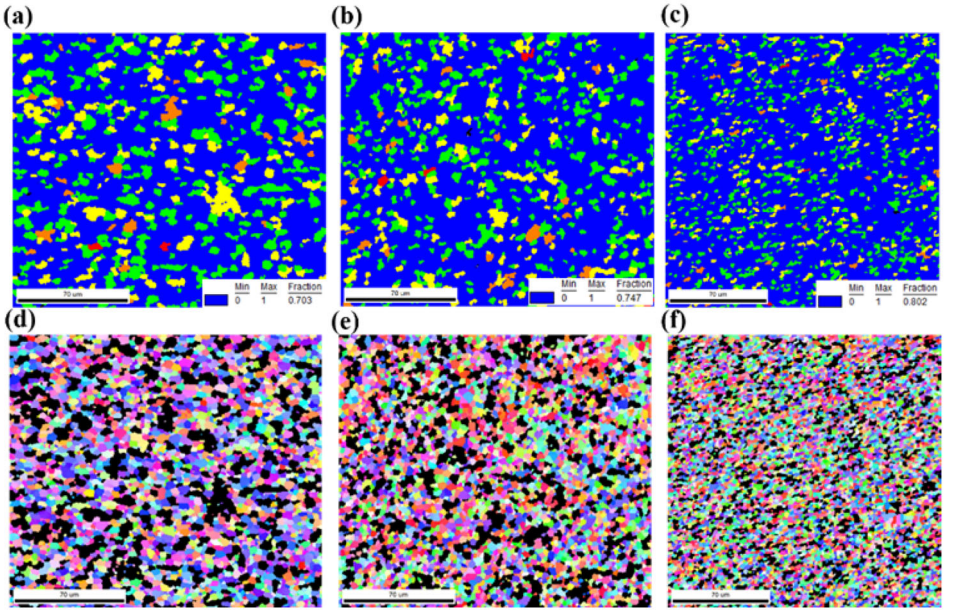


Figure 5. GOS map of welded samples with ω of (a) 600 rpm (b) 900 rpm (c) 1200 rpm at a constant v of 100 mm/Min, PD of 0.1 mm, and its IPF map of welded samples with ω of (d) 600 rpm (e) 900 rpm (f) 1200 rpm at a constant v of 100 mm/min, PD of 0.1 mm.

IPF maps in Figure 5 (d–f). Thus, it is confirmed that DRXed microstructures have been found in SZ.

Microstructure emerges in FSWed areas of aluminium alloy by continuous dynamic recrystallization (CDRX), with dynamic recovery, recrystallization, and grain development acting as driving factors [41]. Mechanical deformation and heat in FSW increase the amount of dislocations in the original microstructure of the base material. The microstructure was restored as the process progressed by annihilation and dislocation rearrangement. Recovery occurs in materials with high stacking fault energy, such as aluminium, where dislocations may migrate from one plane to another through cross slip and climb [41]. After a substantial strain, new grains emerge as a result of the impingement of wavy borders of original grains, which leads to the subdivision of fibrous grains into crystallites bordered by high angle grain boundaries (HAGBs). CDRX involves the formation of stable three-dimensional (3D) arrays of strain-induced low angle grain boundaries (LAGBs) due to dislocation rearrangement, followed by their gradual transformation to HAGBs [40, 42]. The progressive evolution of the subgrains occurs continuously and ultimately leads to their conversion to fine grains [40]. The values of HAGBs and LAGBs of SZ of different samples have been shown in Figure 6. Very low angle grain boundaries (VLAGBs) have their misorientation angle below 5°. LAGBs have misorientation angles in the range of 5°–15°, and HAGBs have misorientation angles greater than 15°. The values of HAGBs and LAGBs were 76.7% and 16.7%, 79.4% and 13.3%, 83.3% and 11.4% with ω of 600 rpm 900 rpm 1200 rpm, respectively. Since all the samples in SZ have nearly 80% HAGBs, it revealed the recovery and recrystallization of microstructures [41].

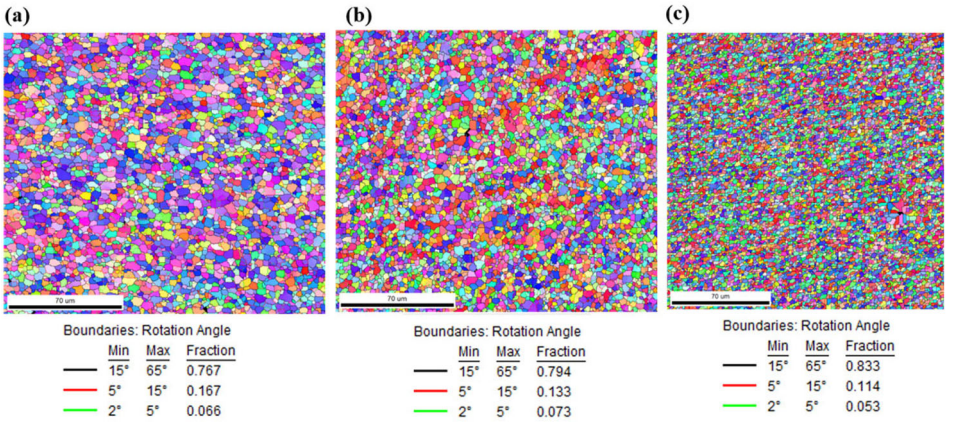


Figure 6. HAGBs, LAGBs of welded samples with ω of (a) 600 rpm (b) 900 rpm (c) 1200 rpm at a constant v of 100 mm/Min, PD of 0.1 mm.

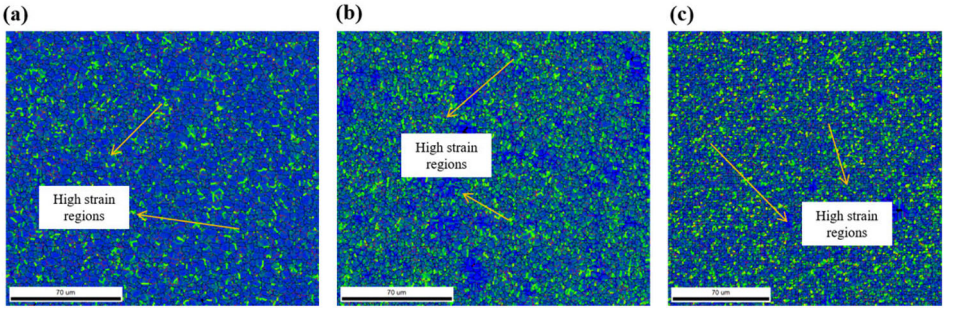


Figure 7. KAM map of welded samples with ω of (a) 600 rpm (b) 900 rpm (c) 1200 rpm at a constant v of 100 mm/Min, PD of 0.1 mm.

A greater proportion of HAGBs can be attributed to a comparatively larger fraction of recrystallization. At a ω of 1200 rpm, the fraction of HAGBs increased at SZ in comparison to 900, and 600 rpm as depicted in Figure 6. The reason can be attributed to more material deformation in SZ obtained with ω of 1200 rpm, generating a large amount of strain and dislocations. According to the literature, CDRX takes place *via* the direct conversion of LAGBs to HAGBs [43]. This leaves high-strain regions near the grain boundaries which can be seen from Kernel Average Misorientation (KAM) map in Figure 7.

The greenish, yellow regions denote the high strain area which is usually more in CDRX[44]. Another distinguishing characteristic of CDRX is the presence of a well-defined subgrain structure in the remnants of unrecrystallized grains which can be seen from the GOS plots and the corresponding IPF plots with VLAGBs, LAGBs, and HAGBs in Figure 6. Hence, from above mentioned results, it is concluded that the grain structure formation is due to CDRX during the FSW of AA1100.

3.1.3. Tensile strength

To prepare the tensile specimen for testing, the welded samples were cut perpendicular to the welding direction using a CNC wire-cut EDM machine (Elecktra, Maxicut

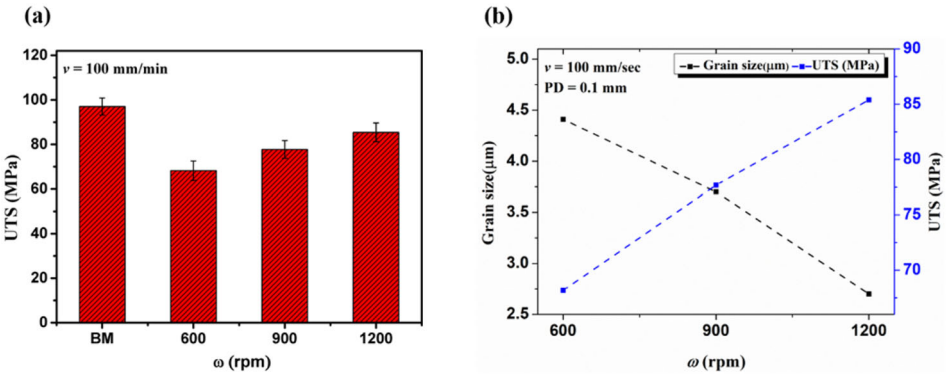


Figure 8. (a) Tensile strength of BM and welded samples with different values of ω (b) combined plot for grain size and UTS with different values of ω.

523), following the guidelines outlined in the ASTM E8 standard [45]. The actual tensile testing was performed using a universal tensile testing machine (Instron, 8862) with a capacity of 100 kN. The testing was carried out at a constant crosshead speed of 1 mm/min.

Figure 8 (a) displays the tensile strength results for both the welded samples and the base material (BM). The tensile strength of the base material was measured to be 97.01 ∓ 3.8 MPa. For the different values of ω, namely 600 rpm, 900 rpm, and 1200 rpm, the corresponding values of tensile strength were observed to be 68.17 ∓ 4.4, 77.69 ∓ 3.9, and 85.39 ∓ 4.2 MPa, respectively after a calculation of standard deviation from 3 tests for each sample. It is worth noting that the welding efficiency at 600 rpm was lower, specifically 70.27%, due to the presence of defects in the weld. However, the welding efficiency for 900 and 1200 rpm was 80.08% and 88.02%, respectively. The welding efficiency increases with an increase in ω because of the grain refinement.

3.2. Simulation outcomes

Each simulation file was created independently, incorporating different process parameters, pin profiles, etc., and run by activating the cluster computing (using batch license) with the command comsolbatch .inputfile < input file path> -outputfile < output file path>.

3.2.1. Validation of the model

The developed model has been validated with peak temperature results that were found experimentally as shown in Figure 9. The temperature obtained from the simulation was identified at an identical location to experiments for all the process conditions.

The percentage errors are observed to be 10.72%, 4.15%, and 1.85% for a ω of 600 rpm, 900 rpm, and 1200 rpm, respectively. The reason behind the differences in experimental and simulated temperature is that the exact temperature was measured from the experiments by the use of thermocouples however, in simulation, the

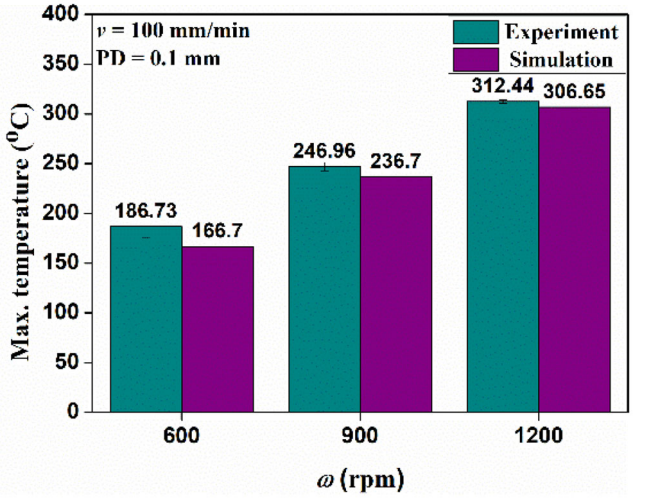


Figure 9. Validation of simulation results with experimental data at different values of ω.

Table 5. Comparison of maximum temperature between the temperature-dependent and constant yield strength of the material (AA1100).

Sl.no.	ω (rpm)	Y (T) Max. temperature (^{0}C)	Y (C) Max. temperature (^{0}C)	Experimentally obtained max. temperature (E_{max}) (^{0}C)	% error between Y (T) and E_{max}	% error between Y (C) and E_{max}
1	600	166.7	165.98	186.73	10.72	11.11
2	900	236.7	235.36	246.96	4.15	4.69
3	1200	306.65	306.14	312.44	1.85	2.01

temperature was calculated based on solving various equations, and hence approximated temperature was found. A few tuning parameters were also considered in the model such as friction coefficient, convective heat transfer coefficient, etc., which are being assumed for the current model and are different from the real-life ones for different process conditions. Simulation results are failing to capture the exact phenomena of FSW in lower tool rotation speeds and due to this more error % came in between the simulation results and the experimental results which are lesser in higher values of ω. The rate of heat generation due to friction and plastic deformation decreases as the temperature rises with higher ω since both are a function of flow stress. Due to this reason, thermal stabilization occurs which leads to capturing the correct physics more in the simulations with higher ω. Therefore, it leads to giving correct values of temperature in this model for higher ω.

3.2.2. Importance of considering temperature-dependent yield stress in the model

The maximum temperature was obtained with a temperature-dependent yield value and with a constant yield value (34 MPa), and the results are shown in Table 5. All the values were taken at a constant v of 100 mm/min, and PD of 0.1 mm. The data presented in Table 5 indicates that there is a reduction in the percentage error with the increase of ω in both cases. In the case of considering temperature-dependent yield value, Y(T) compared to constant yield value, Y(C), there is a lower percentage

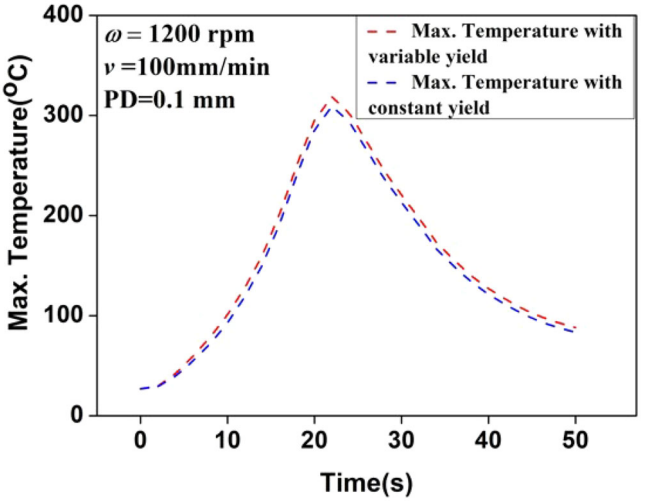


Figure 10. Maximum temperature obtained for temperature-dependent yield and constant yield of AA6061-T6.

error. It is also observed that temperature-dependent yield values have a very less notable effect on the maximum temperature in the case of AA1100. The reason behind this was the material (AA1100) considered for the current study does not have much higher yield strength (34 MPa).

It is anticipated that there must be a significant difference in percentage error of temperature values when the higher yield strength material will be chosen for FSW such as high-strength aluminum, high-strength steel, etc. As a case study, AA6061-T6 material which is having a yield strength of 276 MPa at room temperature was chosen with all similar parameters considered for the present model.

The temperature at the desired point (at 1 mm distance from the shoulder surface and middle of the workpiece) came as 308.29 °C with a constant yield value of AA6061-T6, and 320.18 °C with a temperature-dependent yield value of AA6061-T6, as shown in Figure 10. The temperature-dependent yield considered for the AA6061-T6 material was $24\,^{\circ}C = 276$ MPa, $100\,^{\circ}C = 262$ MPa, $149\,^{\circ}C = 214$ MPa, $204\,^{\circ}C = 103$ MPa, $260\,^{\circ}C = 34$ MPa, $316\,^{\circ}C = 19$ MPa, $371\,^{\circ}C = 12$ MPa. The window sealing action was considered in the model, i.e. above the temperature 371 °C, the yield value was considered as 12 MPa.

3.2.3. Material flow understanding from the temperature distribution

The advancing side (AS) of the aluminum plates during FSW is the side where the direction of tool rotation coincides with the direction of tool traverse whereas the retreating side (RS) is the reverse of the AS. From Figure 11, it was observed that the temperature on AS was more than RS in both the cases of experiments and simulations. The underlying cause is that the deformation and material flow start at AS, thus, the material produces more heat of plastic deformation. Also, more friction is generated since the tangential component of the rotation of the tool and its traverse are the same. In AS, the solid material starts to become semisolid and flows around the tool pin which is plunged into the material. This semisolid material gets

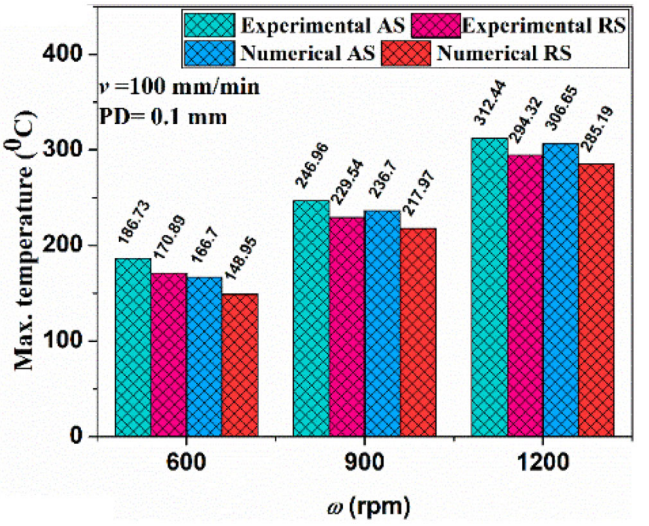


Figure 11. Maximum temperature variation at different ω in As and RS.

consolidated and cooled in RS. Because of this, AS is having more solid nature than RS at any given time or place during FSW. This leads to making more heat, and thus the peak temperature rises. Also, the fact that the tangential component of tool rotation and traverse speeds are opposite in RS resulting less frictional forces. Due to these factors, the temperature on AS is higher than on RS. Differences in temperature between AS and RS during FSW can have several consequences. It might cause uneven heat transfer, and may affect the material flow during welding etc. These variations in temperature and material flow can ultimately impact the properties of the weld joints. There is a variation between 15-19 °C in the maximum temperature of the AS and RS at different ω values keeping v at 100 mm/min and PD at 0.1 mm, as presented in Figure 11.

A cavity is produced when the FSW pin is plunged inside the base metals. The shape of the cavity is the same as the shape of the outer profile of pin. Material flows when the pin in the weld cavity and base metals interact. To understand the material flow, in Figure 12, twelve points were taken at 1 mm away from the periphery of the shoulder in a circle, and temperature values were calculated at those points. When the tool traverses forward to complete the welding, the temperatures are calculated at each point and shown in Figure 12.

In Figure 12, it is observed that temperature is gradually increasing to the front edge of AS. This is happening because of the strong plastic flow of the material which starts from AS. The material starts sheared from the AS. When the tool traverses forward, the material in the leading-edge flows through RS to the trailing edge. Interestingly, the temperature is gradually decreasing towards RS from the leading edge. In this period, the material flows between the FSW tool and the base metals, which is usually cooler. In this zone, hard and soft materials rub against each other, and due to this, the temperature drops. In the RS, the semi-solid material consolidates and cooled. Then, the next tool rotation tries to move the material from RS to AS. In this zone, the plasticized material moves slowly because the rotary tool

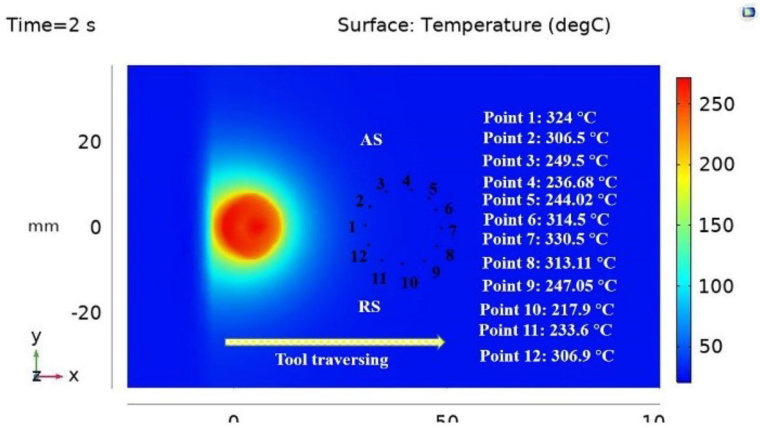


Figure 12. Temperature distribution on the surface.

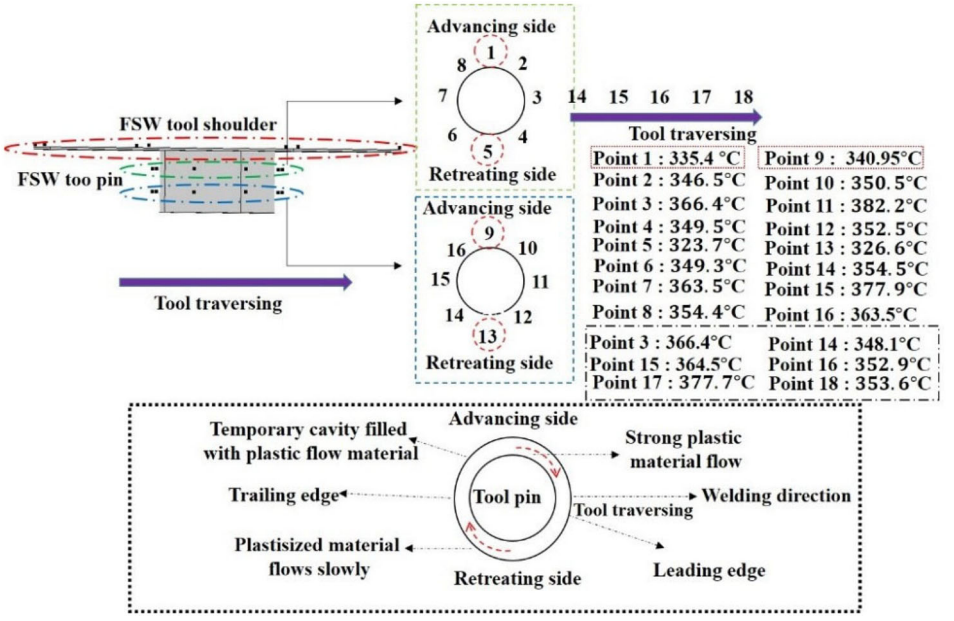


Figure 13. Temperature distribution on the pin periphery.

squeezes the material and pushes it back slowly toward the AS. Thus, the temperature is rising towards the trailing edge. A temporary cavity formation occurs on AS. The plasticized material is extruded into the cavity and filled therein. Hence the temperature decreases from the trailing edge to the advancing side zone as shown in Figure 12.

In Figure 13, eight points were taken in 1 mm, and 2 mm inside the workpiece respectively but around the tool pin. The same temperature pattern was observed there also for the cylindrical pin. This model can help to understand the material flow in detail for various tool pin configurations. A small graphical image was shown on material flow based on the observations in Figure 13.

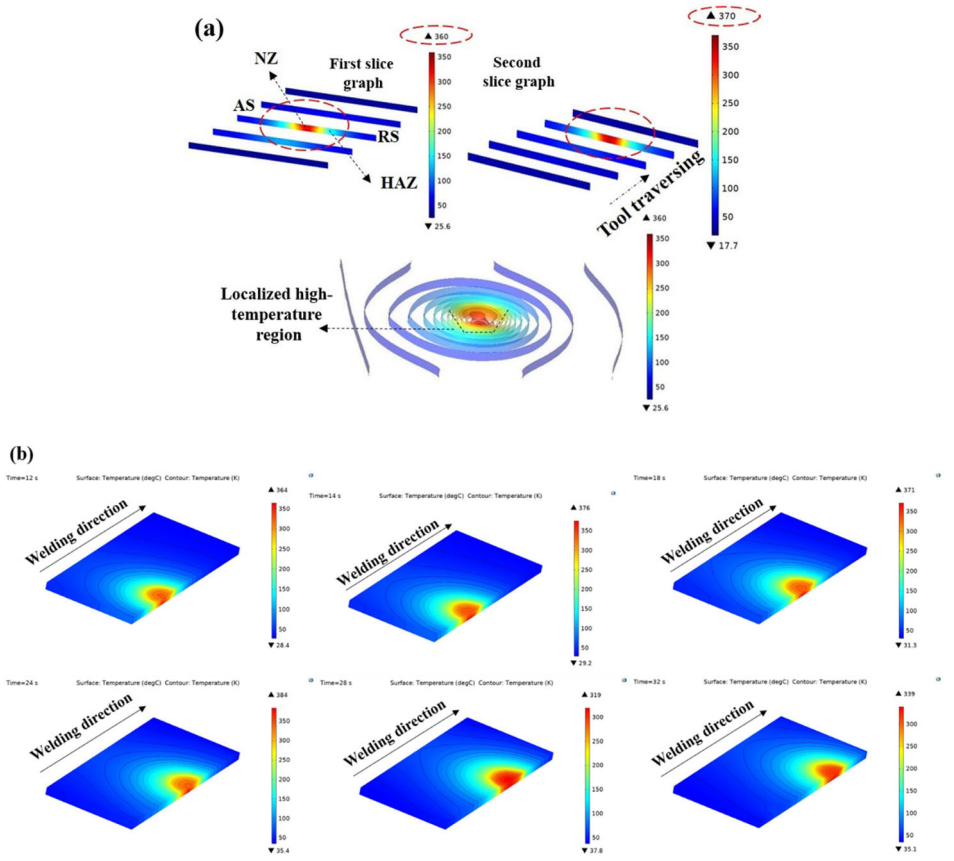


Figure 14. (a) Variation in heat profiles (b) simulated temperature field in the vicinity of FSW tool (sliced along the seaming interface).

In Figures 14 (a and b), it was seen that the heat envelope is changing with the traversing of the tool, which means its heat content is also changing. From the first slice graph in Figure 14 (a), the maximum temperature is seen as 360 °C but in the second slice figure, the maximum temperature is observed as 370 °C. Figure 11 (b) shows the temperature field at 12, 14, 18, 24, 28, and 32 s in the vicinity of the FSW tool. A localized high-temperature region is present in the vicinity of the tool in Figure 14 (b). The simulated maximum temperatures for AA1100 are lower than its melting point, which provides context since FSW is a solid-state joining technique. A conclusion can be made that if the extent of the peak temperature is high, then it will affect the weld structure which will have an impact on the microstructure and grain orientation of the weld at that portion. In Figure 13, more five points were taken in the forward direction beside point 3, and the maximum temperature was found in all those points. The maximum temperature was found to be varied which will of course affect the microstructure, texture etc. Because of this change, material stress value is observed. The distribution of von Mises stress can give an insight to material stirring which is depicted in Figure 15. A similar comment has been found in the reference [46].

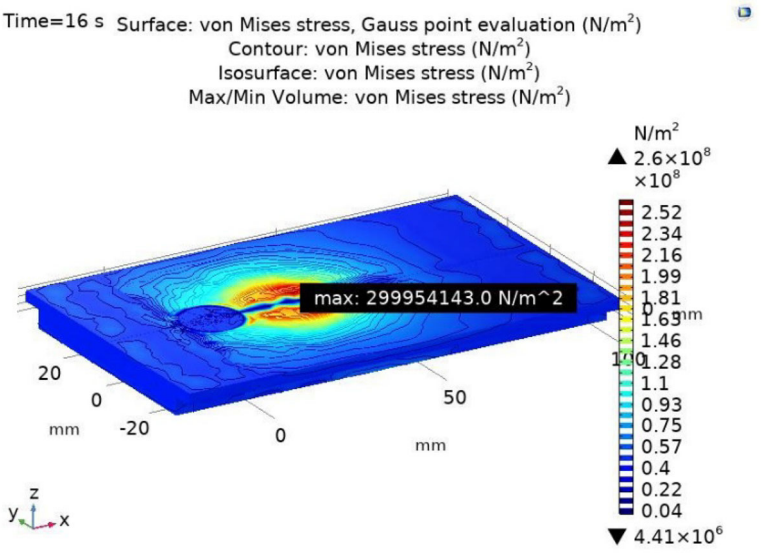


Figure 15. von Mises stress distribution in FSWed plate.

The von Mises stress in FSW, accounts for the material expansion during the heating of weld plates, and then contraction at the time of the cooling of the weld plates. Also, the rotation and forward motion of the tool adds stress to the weld because of the mechanical constraint of workpieces by fixtures and the backing plates. The proposed model is primarily intended to predict stress distributions over the whole FSW process. Figure 15 depicts the von Mises stress plot (N/m^2) at a time point of $t = 16$ s (considering the process parameters as $\omega = 1200$ rpm, $v = 100$ mm/min, PD $=$ 0.1 mm) from the start of the process. A high-stress layer is observed in the center of the workpieces. The cylindrical high-stress layer progressively is disappearing since the metal was heated up by the generated heat due to plastic deformation at the shoulder–pin juncture. Dissimilarity in temperature in AS and RS dictates an eloquent effect on von Mises stress and as a consequence on the weld formation. It was found that the stress on the AS and RS (at a 1 mm distance from the shoulder surface and middle of the workpiece) were 1.5272E8 (N/m^2) and 1.4283E8 (N/m^2). The reason for higher stress in AS is attributed to the maximum temperature obtained in AS during welding than the RS.

3.2.4. FSW with polygonal pin profiles

In FSW, pin profile is an essential characteristic that affects the grain size, mixing of material, weld temperature, plastic deformation, weld quality, etc. A significant amount of material is extruded in the vicinity of the pin in FSW, except the topmost surface, where deformation is more difficult due to contact with the FSW tool shoulder.

In addition to process variables such as ω, v, and PD, tool pin shape has a significant impact on the strain rate in the SZ. As a result, tool pin shape is an important factor influencing the deformation mechanics of FSW process [47]. The model was validated for a cylindrical pin profile; however, the present research takes into

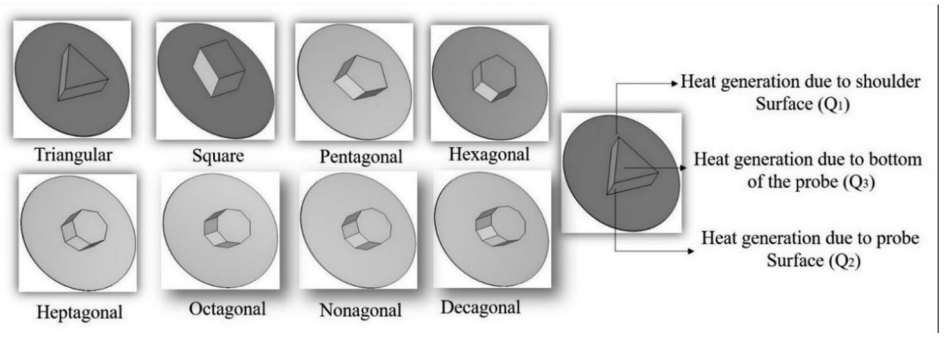


Figure 16. Schematic of various profiles of pin.

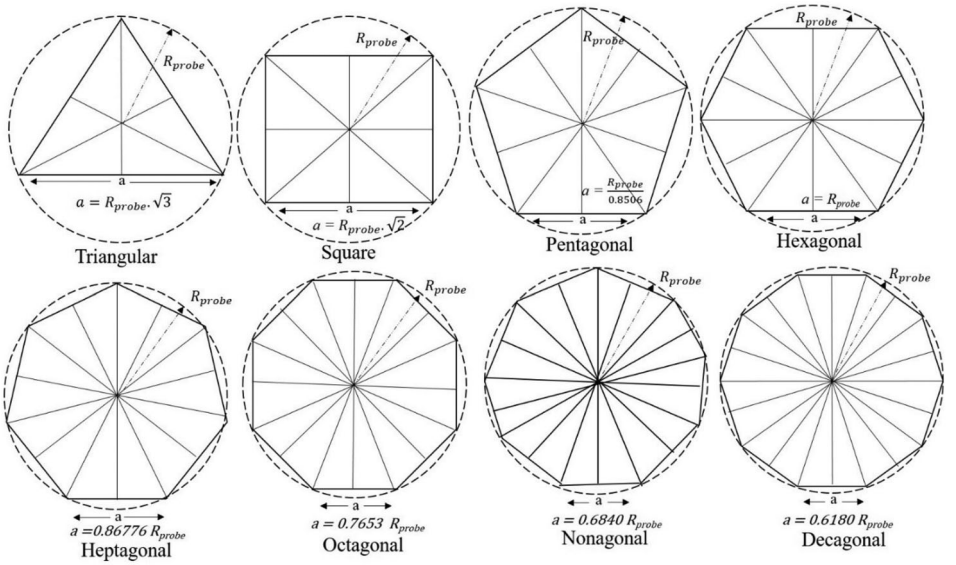


Figure 17. Schematic of R_{probe} for various profiles of pin.

account diverse polygonal pin shapes. Those are: triangular to decagonal pin profiles as shown in Figure 16. For various geometries, the pin diameter across the corner of the probe pin was also kept constant (radius = 2.5 mm) as shown in Figure 17.

Table 6 represents the calculation of the side of the edge for different pin profiles. The purpose of this was to ensure the consistency of the outcomes and avoid any alterations in the results caused by variations in the swept diameter. The geometries are created in SOLIDWORKS and imported in COMSOL multiphysics *via* LiveLink™ for SOLIDWORKS®, as shown in Figure 17. This procedure will help the researchers to import different pin profiles to work further. The equations considered for heat generation due to the shoulder surface, pin periphery, and bottom surface of the pin for different pin profile simulations are calculated accurately and presented in Table 7. These equations can be used to analyze further the relationship between the

Table 6. Calculation of the side of the edges of the pin profiles.

Pin profiles	Pin edge (a)
Triangular	$a = R_{probe} \cdot \sqrt{3}$
Square	$a = R_{probe} \cdot \sqrt{2}$
Pentagonal	$a = \frac{R_{probe}}{0.8506}$
Hexagonal	$a = R_{probe}$
Heptagonal	$a = 0.86776\ R_{probe}$
Octagonal	$a = 0.7653\ R_{probe}$
Nonagonal	$a = 0.6840\ R_{probe}$
Decagon	$a = 0.6180\ R_{probe}$

Table 7. Equations used for different pin profiles.

Triangular Pin Profile	$Q_1 = \frac{2}{3}.\pi.\omega.\tau_{contact}.R^3_{shoulder} - \frac{2.\pi.\omega.\tau_{contact}.a^3}{9\sqrt{3}}$
	$Q_2 = \frac{1}{2}.\omega.\tau_{contact}.a^2.H_{probe} + \frac{2.\pi.\omega.\tau_{contact}.a^3}{9\sqrt{3}}$
	$Q_3 = \frac{2.\pi.\omega.\tau_{contact}.a^3}{9\sqrt{3}}$
Square Pin Profile	$Q_1 = \frac{2}{3}.\pi.\omega.\tau_{contact}.R^3_{shoulder} - \frac{\pi.\omega.\tau_{contact}.a^3}{3\sqrt{2}}$
	$Q_2 = \omega.\tau_{contact}.a^2 H_{probe}$
	$Q_3 = \frac{\pi.\omega.\tau_{contact}.a^3}{9\sqrt{3}}$
Pentagonal Pin Profile	$Q_1 = \frac{2}{3}.\pi.\omega.\tau_{contact}.R^3_{shoulder} - \frac{2\times0.6155.\pi.\omega.\tau_{contact}.a^3}{3}$
	$Q_2 = 1.8088.\omega.\tau_{contact}.a^2.H_{probe}$
	$Q_3 = \frac{2\times0.6155.\pi.\omega.\tau_{contact}.a^3}{3}$
Hexagonal Pin Profile	$Q_1 = \frac{2}{3}.\pi.\omega.\tau_{contact}.R^3_{shoulder} - \frac{2.\pi.\omega.\tau_{contact}.a^3}{3}$
	$Q_2 = 3.\omega.\tau_{contact}.a^2 H_{probe}$
	$Q_3 = \frac{2\pi.\omega.\tau_{contact}.a^3}{3}$
Heptagonal Pin Profile	$Q_1 = \frac{2}{3}.\pi.\omega.\tau_{contact}.R^3_{shoulder} - \frac{2.\pi.\omega.\tau_{contact}.a^3\times1.5303}{3}$
	$Q_2 = 4.64802\omega.\tau_{contact}.a^2 H_{probe}$
	$Q_3 = \frac{2\pi.\omega.\tau_{contact}.a^3\times1.5303}{3}$
Octagonal Pin Profile	$Q_1 = \frac{2}{3}.\pi.\omega.\tau_{contact}.R^3_{shoulder} - \frac{2.\pi.\omega.\tau_{contact}.a^3\times2.2310}{3}$
	$Q_2 = 6.8296\omega.\tau_{contact}.a^2 H_{probe}$
	$Q_3 = \frac{2\pi.\omega.\tau_{contact}.a^3\times2.2310}{3}$
Nonagonal Pin Profile	$Q_1 = \frac{2}{3}.\pi.\omega.\tau_{contact}.R^3_{shoulder} - \frac{2.\pi.\omega.\tau_{contact}.a^3\times3.1248}{3}$
	$Q_2 = 9.6183\omega.\tau_{contact}.a^2 H_{probe}$
	$Q_3 = \frac{2\pi.\omega.\tau_{contact}.a^3\times3.1248}{3}$
Decagonal Pin Profile	$Q_1 = \frac{2}{3}.\pi.\omega.\tau_{contact}.R^3_{shoulder} - \frac{2.\pi.\omega.\tau_{contact}.a^3\times4.2367}{3}$
	$Q_2 = 13.0916\omega.\tau_{cotact}.a^2 H_{probe}$
	$Q_3 = \frac{2\pi.\omega.\tau_{contact}.a^3\times4.2367}{3}$

Here, H_{probe} denotes pin height which is 2.8 mm for all the pin profiles, $\tau_{contact}$ is frictional shear stress, a is the side of the pin, R_{probe} *the* radius of pin profiles.

temperature distribution and weld quality i.e. ultimate tensile strength using polygonal pins.

Figure 18 illustrates the maximum temperature distribution (obtained at 1 mm distance from the tip of the tool shoulder and just exactly in the middle of the workpiece) with the conventional triangular to decagonal pin profiles. An increase in maximum temperature from triangle to decagonal pin profile is caused by higher friction heating on and under the vertical pin surfaces as the number of pin sides increases.

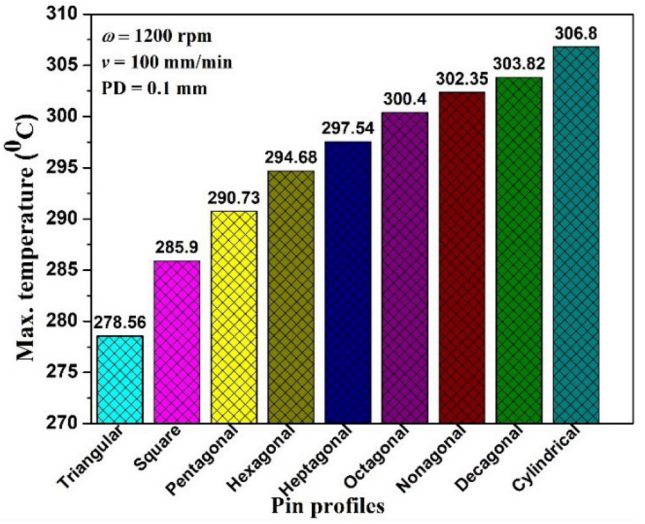


Figure 18. Variation of maximum temperature for various profiles of pin.

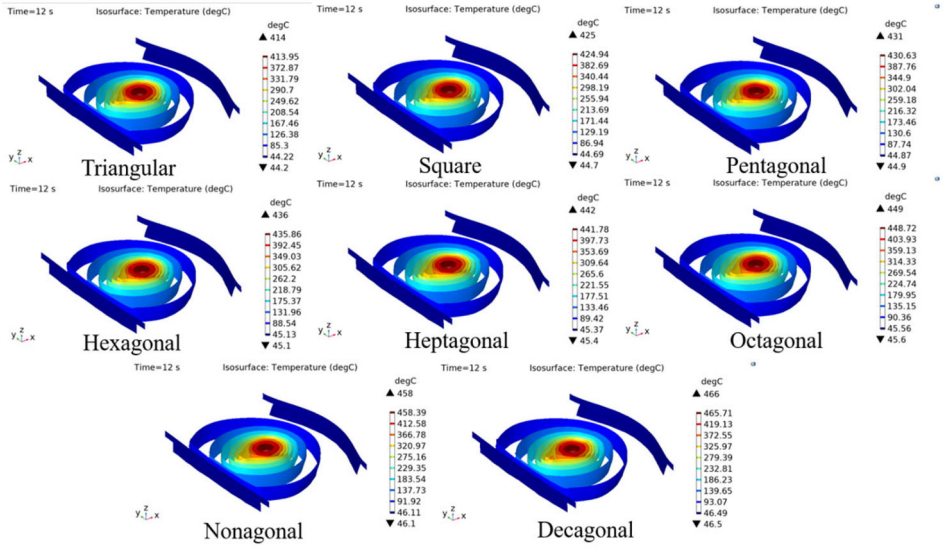


Figure 19. Isothermal contours of different pin profiles.

Figure 19 depicts the isothermal contour plot of eight different regular polygonal pins. More number of pin sides lead to more pin-to-work contact, which results in more frictional heat production, especially on the vertical surfaces of the pin. As a result, this leads to an increase in the maximum temperature. With the further increase in the number of faces, the increase in total heat generation is very small which can be seen from octagonal to decagonal pin profiles. These findings are well supported by prior studies where four-pin shapes (Triangular, Square, Pentagonal, Hexagonal) were considered to perform the FSW [48]. When the number of sides of pin increases, the heat generation is gradually approaching the heat generation caused by the cylindrical shape pin. The cylindrical pin has a more contact surface with the

workpiece at the lateral surface in comparison to polygonal pin profiles. Increasing the contact area increases the amount of frictional heat which leads to more heat generation. This research is equivalent to the previously published literature [49–51].

3.2.5. Effect of different backing plates on the peak temperature

FSWed joints have several essential characteristics such as ω, v, the diameter of the tool shoulder, forge force, etc. In FSW, the thermal diffusivity of backing plates also plays a key character to produce a good weld. The contact gap conductance and the rate of heat flow through the backing plate itself were influenced by the appropriate selection of backing plate materials which can have a big effect on how the temperature is distributed inside the workpiece. The capacity of the material to transmit heat from the heated area to the cool portion is termed as thermal diffusivity.

A variety of materials, for example, mild steel, pure copper, aluminum alloys, medium carbon steel, granite, floor ceramic tiles, rusty steel, Ti-6-4, stainless steel, etc., may be utilized as a backing plate. Some of them are considered for the present study. These are Tungsten, Titanium, Mild steel, Granite, Copper, Cast iron, and Al 6063-T83. All the results mentioned in the above sections for both experiments and simulations have been captured by taking cast iron as the backing plate material. The variation of maximum temperature (obtained at 1 mm distance from the tip of the tool shoulder and just exactly in the middle of the workpiece) concerning different backing plates is shown in Figure 20 (b). In all cases, the process parameters were kept constant ($\omega =$ 1200 rpm, $v = 100$ mm/min, PD = 0.1 mm).

The temperature and welding time curves are shown in Figure 20 (a) which indicate that the predicted peak temperature in simulation with different backing plates depends on its thermal conductivity and thermal diffusivity. The calculation of thermal diffusivity for different backing plates is given in Table 8. The aluminum 6063 –T83, cast iron, Copper, Granite, and Mild steel showed peak temperatures (at 1 mm distance from the shoulder surface and middle of workpieces) of 241.43, 306.66, 198.13, 381.64, 299.95, 350.03, and 249.71 °C, respectively. The temperature rises as the thermal conductivity of the backing plate decrease, as shown in Figure 20. Low thermal conductivity will lead to lesser heat loss by conduction to the FSW machine.

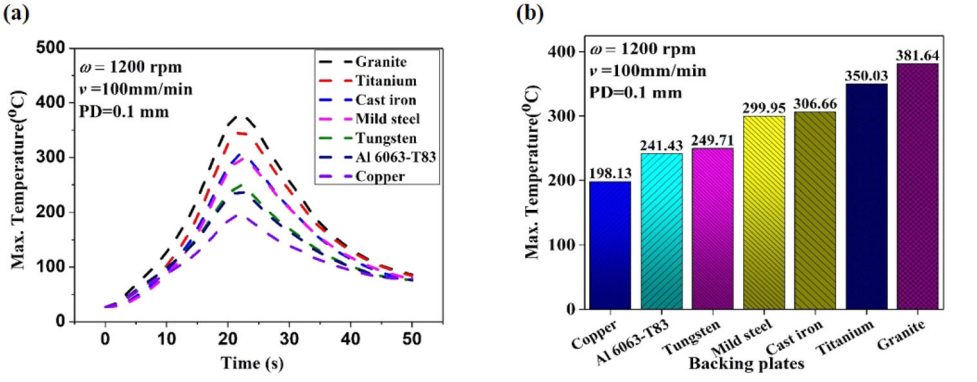


Figure 20. (a) Variation of temperature with time for different backing plates (b) effect of different backing plates on maximum temperature during FSW.

Table 8. Calculation of thermal diffusivity for different backing plates.

Different backing plates	ρ [kg/m3]	k [W/(m*K)]	C_p [J/(kg*K)]	Density* Specific heat at constant pressure	Thermal diffusivity (m^2 s^{-1})
Copper	8960	400	385	3449600	11.5955E-05
Tungsten	17800	175	132	2349600	7.44808E-05
Aluminum 6063-T83	2700	201	900	2430000	8.2716E-05
Mild steel	7854	64	472	3707088	1.72642E-05
Cast iron	7000	50	420	2940000	1.70068E-05
Titanium beta-21S	4940	7.5	710	3507400	0.213834E-05
Granite	2600	2.9	2600	2210000	0.131222E-05

It resembles the findings of a few earlier research [52, 53] where high thermal conductivity materials were utilized to decrease the peak temperature. Since the current model is novel in terms of accuracy, and computational time, hence it will help the researchers to work on various backing plates and check the weld quality with different backing plates. It is also evident from the experiments that the use of backing plates has a strong influence on the mechanical properties of weld joints e.g. strength and hardness. Backing plate materials with higher thermal diffusivity, such as copper and aluminum 6063-T83, result in a higher heat extraction rate. The heat transfer rate is reduced in materials with lower thermal diffusivity, e.g. granite. The diffusivity exhibits an increasing trend across different materials, starting from Granite, followed by Titanium, Cast iron, Mild steel, Tungsten, Aluminium, and Copper, as calculated in Table 8, which increases the peak temperature.

4. Comparison of FSW model with another approach to check the computational efficiency

A thermo-mechanical nonlinear transient model of FSW is simulated using ABAQUS/Explicit software by coupled eulerian lagrangian (CEL) method. All the dimensions and materials of the workpieces and tool were kept consistent in accordance with the experiments as well as with the COMSOL model described above. The workpiece was considered as an Eulerian body in this model and the tool was considered as a lagrangian body which is rigid. The workpiece was partitioned into two halves. The upper 1 mm domain is the void region and the below 3 mm domain is the material-assigned region. Johnson-Cook (JC) material model was used to calculate the flow stress of material which is expressed by Equation (14).

$$\sigma = [A + B\varepsilon^n]\left[1 + C\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0}\right)\right]\left[1 - \left(\frac{T - T_{room}}{T_{melt} - T_{room}}\right)^m\right] \tag{14}$$

where A is the initial yield strength of the material, B is the hardening modulus and C is strain-rate dependency. T_{melt}, T_{room}, $\dot{\varepsilon}_0$, $\dot{\varepsilon}$, ε, σ, n, m are the melting point of the workpiece metals, room temperature, reference strain rate, strain-rate, effective strain, flow stress, strain hardening coefficient, and thermal softening coefficient respectively. The values are tabulated in Table 9.

The assumptions, boundary conditions, and contact conditions considered in the model are highlighted in Figure 21. The temperature contour plot was found from

Table 9. JC Constants for AA1100.

A (MPa)	B (MPa)	C	M	n	T_{room} (^{0}C)	T_{melt} (^{0}C)
34	180	0.001	0.185	0.2	24	660

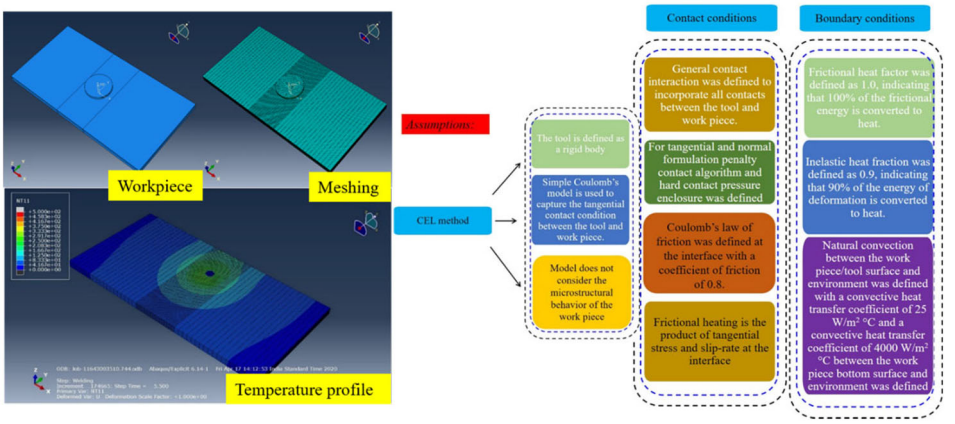


Figure 21. FSW model in ABAQUS.

the ABAQUS model. In the FSW model built in ABAQUS software, it took 2 min 44 sec to complete 2.024×10^{-4} step time. It implies that to complete 15 timesteps in the model, it will take more than four months which is computationally very expensive. Then mass scaling of 10^6 was performed in the model to reduce this computational time. However, the CEL model took 25 h to finish the simulation which is 3 h in the COMSOL simulated FEM model to predict the temperatures. There are many ways to develop a three-dimensional thermo-mechanical model to simulate the FSW process. In the present work, two ways are explained for building the model of FSW which can predict the same output responses i.e. temperature. CEL is also a well-known method to simulate FSW. An attempt has been made to compare the computational time of the developed model with respect to the CEL. The CEL model is inline with the published literature [28, 54]. The developed model has been optimized to minimize the computational time.

Researchers can make mathematical model order reduction (MOR) as a reliable modeling approach, which is formal, numerically robust, and can be fully automated. By using LiveLink^{TM} for MATLAB, from this COMSOL model, researchers can extract the spatially discretized frequency-domain model and combine algorithms to compute a compact model with only a few degrees of freedom (DOFs). This will help to reduce the solution time for a harmonic solution by several orders of magnitude, with a negligible loss of accuracy. This approach can be used in real-time while making the digital twin of the FSW process.

5. Conclusion

The effect of different backing plate materials and pin profiles on the temperature distribution of AA1100 plates is investigated using a multiphysics 3D model of FSW.

This computationally efficient model is validated with experimental temperature results and can be applied to real-time applications in a variety of industries. As a result of this work, the industries can choose the proper combination of backing plate material and tool pin configuration to produce a good weld. Furthermore, in the age of Industry 4.0, the developed model for its convenience to solve FSW in one platform, i.e. in COMSOL will also help the industries to make a digital twin of FSW for predicting the quality of welding. The digital twin may be used to build the real-time 3D temperature field and to access the temperature field as digital data, enabling more flexible control of the FSW process.

The following are several important conclusions:

1. It was found that with an increase in ω i.e, in 1200 rpm, there was a noticeable improvement in the surface quality of the weld. The higher ω value resulted in a reduction in the occurrence of flashes.
2. It was also observed that with an increase in ω, there was a notable decrease in the grain size of the weld. This reduction in grain size is significant as it directly contributes to an increase in the weld strength. In particular, the maximum joint efficiency of 88.02% was achieved at ω of 1200 rpm, with a tool traversing speed of 100 mm/min and plunge depth of 0.1 mm.
3. It was found that the temperature-dependent yield strength of the workpiece material had a significant effect on the maximum temperature. Therefore, a temperature-dependent yield value must be taken into consideration when modeling the FSW. The temperature values help to understand the material flow.
4. It was observed that a reduction in the peak temperature was found while using high thermal conductivity and thermal diffusivity backing plate. For welding of AA1100 material, cast iron as a backing plate is suitable.
5. Having more sides of pin increases the total pin-to-work contact area, which causes a more frictional heat production rate, especially on the lateral surfaces of the pin. Consequently, this leads to an increase in the maximum temperature. The precise heat generation equations for various polygonal pin profiles were developed which can help researchers to study the temperature distribution for different material combinations.
6. The computational efficiency of the present model is demonstrated by comparison with models from the existing literature.

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The author obliged all the rules regarding ethics in publication.

Consent to participate

All the authors consent regarding the data provided in the text as well as the order of authorship.

Consent for publication

All the authors consent regarding publication immediately after acceptance.

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All data and material provided in the paper are not copied from other sources.

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